

# Data Handling: Import, Cleaning and Visualisation

## Exercise 4: Data Import and Data Preparation/Manipulation

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### Exercise A

Read Section 11.-11.4 in <https://r4ds.had.co.nz/data-import.html> and solve the problem posed in Section 11.4.2 (“Problems”) by going step-by-step through the instructions. Hint: You will need to have the **readr** package installed and loaded (should be also installed/loaded with **tidyverse**).

### Exercise B

1. Download the file **airquality\_data.xlsx** posted on Canvas and save it in your **r\_course/data** folder.
2. The file is a dataset that contains the New York air quality measurements. Familiarize with the data using **print**. Do you think the dataset has problems? (Hint: look at the **type** of each column).
3. Fix the problem identified in point 2. Save the new dataset as **airquality\_data\_fixed.xlsx** using **write\_xlsx** from **tidyverse/writexl**.
4. Write your solution down in an R-Script and document in this script why certain problems occur and how you solved them (using comments).

### Exercise C

1. Visit the EU Open Data Portal on COVID cases.
2. Download the “COVID-19 Cases Worldwide” data in JSON format. However, do not download these data manually (i.e., do not click-and-save). Instead, load the data directly into RStudio from the web. Hint 1: consider the **jsonlite** package. Hint2: the direct link to the JSON file (<https://opendata.ecdc.europa.eu/covid19/casedistribution/json/>) may be helpful.
3. Reformat the JSON data that you have imported into RStudio to a rectangular format (such that each record is represented as a row and each field as a column).